

Amazon EC2 instance-Task

Task 1: Launch an EC2 using Amazon Linux 2 and Install Apache via User Data

Step 1: Launch Instance

1. Open **EC2 Console** → **Instances** → **Launch Instance**
2. **Name:** amazon-linux-apache
3. **AMI:** Amazon Linux 2 (Free-tier eligible)
4. **Instance Type:** t2.micro
5. **Key Pair:** Choose or create
6. **Network:** Default VPC
7. **Subnet:** Choose any
8. **Auto-assign Public IP:** Enable
9. **Security Group:**
 - Allowed SSH (22)
 - Allowed HTTP (80)

Step 2: Added bash script User Data to Install Apache

While launching instance in **Advanced Options** then there is a box **User Data** and i wrote below script:

```
#!/bin/bash
```

```
yum update -y
```

```
yum install httpd -y
```

```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "<h1>Apache Installed Successfully on Amazon Linux 2</h1>" >  
/var/www/html/index.html
```

Step 3: Launch Instance

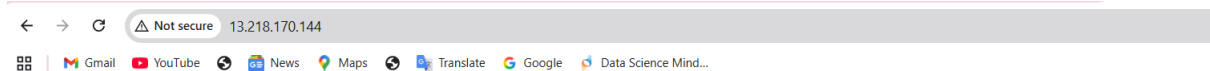
After the instance is running → copy Public IP → open in browser:

Now we can see "Apache Installed Successfully" in web page.

```
Arjuman@Arjuman MINGW64 ~/Downloads (master)
$ ssh -i "Testing.pem" ec2-user@ec2-13-218-170-144.compute-1.amazonaws.com
The authenticity of host 'ec2-13-218-170-144.compute-1.amazonaws.com (13.218.170.144)' can't be established.
ED25519 key fingerprint is SHA256:RK0XmQLf824np/6z3Jahy3JXQzgQ8Wqb5WiigWvuFE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-218-170-144.compute-1.amazonaws.com' (ED25519) to the list of known hosts.

#####
Amazon Linux 2023
#####
V- -> https://aws.amazon.com/linux/amazon-linux-2023
#####

[ec2-user@ip-172-31-31-209 ~]$ sudo systemctl status
● ip-172-31-31-209.ec2.internal
State: running
Units: 291 loaded (incl. loaded aliases)
Jobs: 0 queued
Failed: 0 units
Since: Wed 2025-11-19 09:47:05 UTC; 13min ago
systemd: 252.23-8.amzn2023
CGroup: /
└─init.scope
├─┬1 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
├─system.slice
├─┬acpid.service
├─┬2312 /usr/bin/systemd-inhibit --what=handle-suspend-key:handle-hibernate-key --who=noah "--why=acpid instead" --mode=block /usr/sbin/acpid -f
├─┬2344 /usr/sbin/acpid -f
├─amazon-ssm-agent.service
├─┬2476 /usr/bin/amazon-ssm-agent
├─atd.service
├─┬2483 /usr/sbin/atd -f
├─auditd.service
├─┬2134 /sbin/auditd
├─chronyd.service
├─┬2508 /usr/sbin/chronyd -F 2
├─dbus-broker.service
├─┬2320 /usr/bin/dbus-broker-launch --scope system --audit
├─┬2328 dbus-broker --log 4 --controller 9 --machine-id ec26c2063cefc4f3c1902032e906c966 --max-bytes 536870912 --max-fds 4096 --max-matches 16384 --audit
├─gssproxy.service
├─┬2348 /usr/sbin/gssproxy -D
├─httpd.service
├─┬1748 /usr/sbin/httpd -DFOREGROUND
├─┬3877 /usr/sbin/httpd -DFOREGROUND
├─┬3886 /usr/sbin/httpd -DFOREGROUND
├─┬3890 /usr/sbin/httpd -DFOREGROUND
├─┬3913 /usr/sbin/httpd -DFOREGROUND
├─libstoragemgmt.service
├─┬2315 /usr/bin/lsm -d
├─sshd.service
├─┬2480 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
└─system-getty.slice
```



Apache Installed Successfully on Amazon Linux 2

2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

```
Arjuman@Arjuman MINGW64 ~/Downloads (master)
$ ssh -i "Testing.pem" ubuntu@ec2-34-224-81-134.compute-1.amazonaws.com
The authenticity of host 'ec2-34-224-81-134.compute-1.amazonaws.com (34.224.81.134)' can't be established.
ED25519 key fingerprint is SHA256:tfdtf/QHA9dbE6A8Mz5ntCCQVQvn3WqNCxX1P2jeuQU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-34-224-81-134.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Nov 19 10:31:45 UTC 2025

System load: 0.0          Processes: 111
Usage of /: 29.6% of 6.71GB Users logged in: 0
Memory usage: 22%        IPv4 address for enX0: 172.31.17.156
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
9 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-17-156:~$ sudo systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Active: active (running) since Wed 2025-11-19 10:25:45 UTC; 8min ago
Docs: man:nginx(8)
Main PID: 1494 (nginx)
Tasks: 2 (limit: 1121)
Memory: 1.8M (peak: 3.7M)
CPU: 16ms
CGroup: /system.slice/nginx.service
└─1494 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
    └─1496 "nginx: worker process"

Nov 19 10:25:45 ip-172-31-17-156 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Nov 19 10:25:45 ip-172-31-17-156 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-17-156:~$
```

Activate Win
Go to Settings to

Launched an EC2 instance

- Selected **Ubuntu** under the OS/AMI section.

2 Added the User Data script

Script you wrote:

```
#!/bin/bash
```

```
apt update -y
```

```
apt install nginx -y
```

```
systemctl start nginx
```

```
systemctl enable nginx
```

3 Connected and verified

- Connected through terminal/EC2 console and confirmed Nginx was running.



Nginx on Ubuntu

3. Launch one Windows server and install Tomcat on Windows. Steps You Performed (Short & Clear)

1 Launched a Windows EC2 instance

- Selected **Windows Server**
- Instance type: **t3.medium**
- Used existing key pair: **Testing.pem**
- Added Security Group rules: **RDP (3389)** and **Tomcat (8080)**

2 Retrieved Windows Administrator password

- Went to **Actions → Security → Get Windows Password**
- Uploaded **Testing.pem**
- Decrypted and copied the password

3 Connected to EC2 using RDP

- Downloaded the **RDP file**
- Logged in using:
 - Username: **Administrator**
 - Decrypted password

4 Installed Java

- Downloaded **Temurin JDK 17 MSI**
- Installed normally (Next → Next)

5 Installed Apache Tomcat

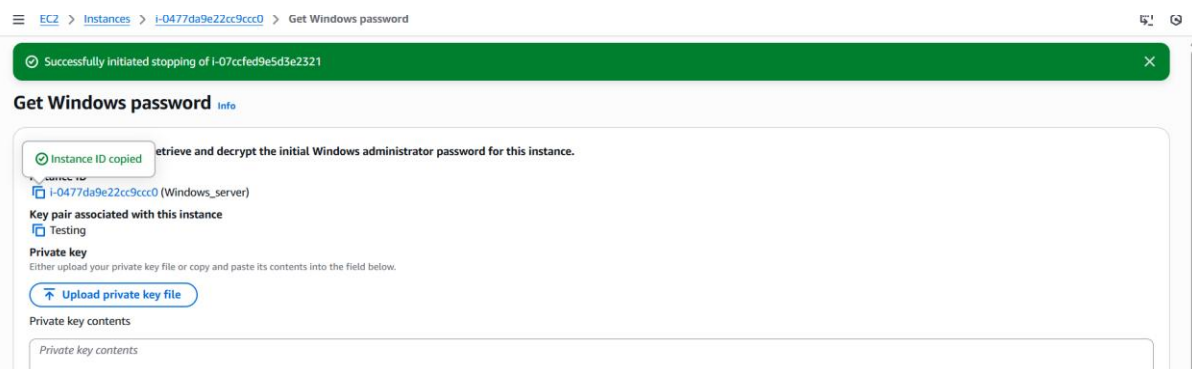
- Downloaded **Tomcat Windows Service Installer (.exe)**
- Installed → Tomcat service started automatically

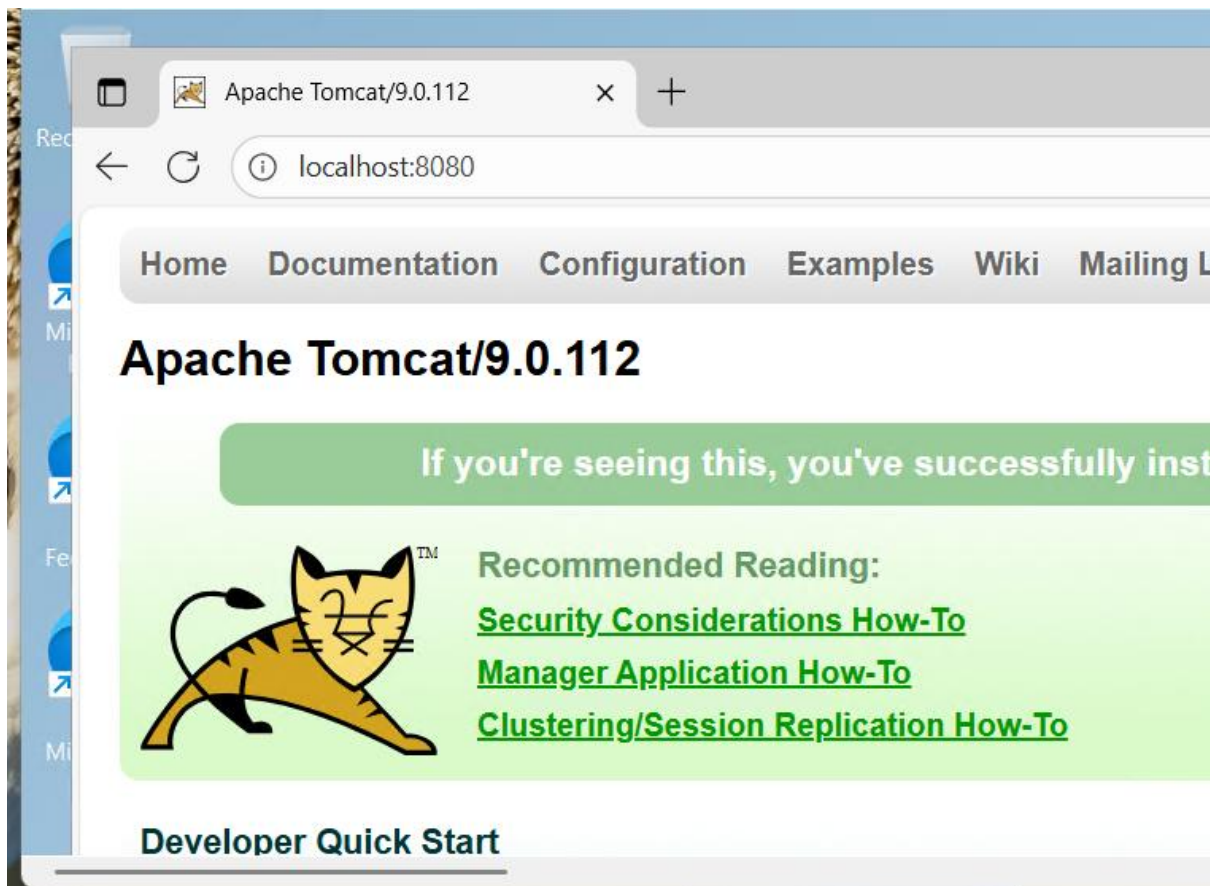
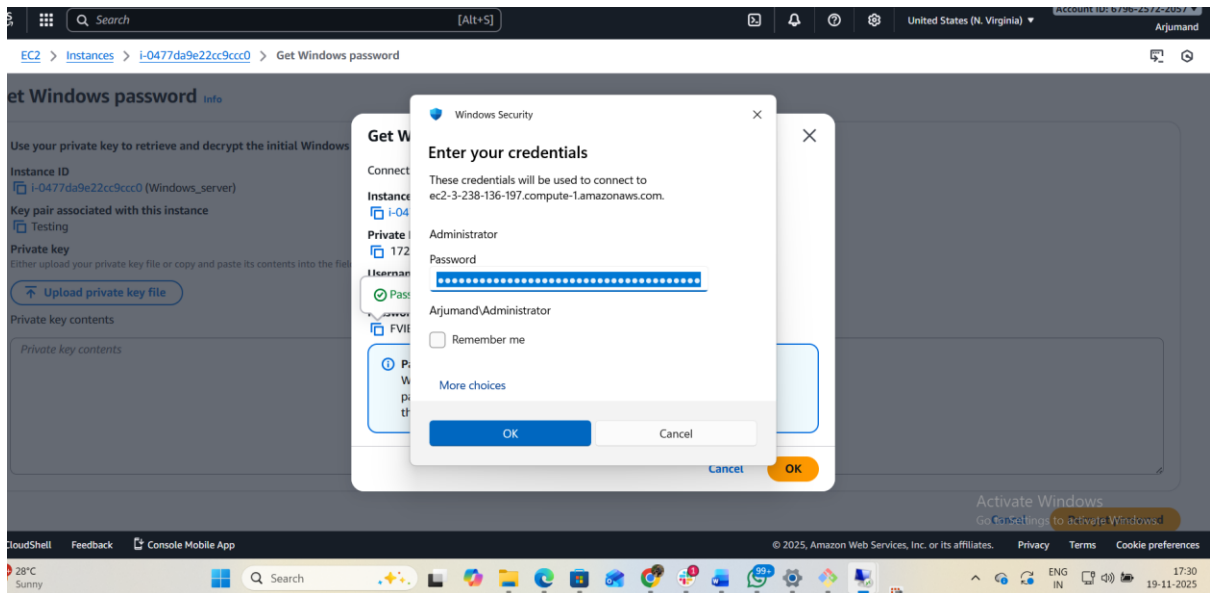
6 Allowed port 8080 in Windows Firewall

- Added inbound rule for **TCP 8080**

7 Verified Tomcat

- Opened browser:
`http://<EC2-public-ip>:8080`
- Saw the **Apache Tomcat welcome page**





4. Take a snapshot of the instance created in Task 1

Create Snapshot (Amazon Linux Apache)

1 Select the Amazon Linux instance

- EC2 → Instances → Click the **Amazon Linux Apache** instance
- Go to **Storage** tab.

2 Open the root volume

- Click the Volume ID (example: **vol-01178b16beeb6b592**).

3 Create snapshot

- **Actions** → **Create snapshot**
- Add a name/description
- Click **Create snapshot**

Snapshot created

- Confirm in EC2 → **Snapshots**

The screenshot displays the AWS Management Console interface. At the top, a green notification banner states: "Successfully created snapshot snap-02b2dc72c620e5032 from volume vol-01178b16beeb6b592. If you need your snapshot to be immediately available consider using Fast Snapshot Restore." Below this, the "Volumes (1)" section is active, showing a table with one volume: vol-01178b16beeb6b592 (gp3, 8 GiB, 3000 IOPS, 125 Throughput). The "Actions" dropdown menu is open, showing "Create snapshot". Below the table, the "Snapshot summary" section shows "0 / 3" recently backed up volumes out of a total of 3 volumes. The "Data Lifecycle Manager default policy for EBS Snapshots status" section indicates that no default policy is set up.

Successfully created snapshot snap-02b2dc72c620e5032 from volume vol-01178b16beeb6b592.
If you need your snapshot to be immediately available consider using Fast Snapshot Restore.

Manage fast snapshot restore

Volumes (1) Info

Saved filter sets
Choose filter set

Volume ID = vol-01178b16beeb6b592

Clear filters

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created
	vol-01178b16beeb6b592	gp3	8 GiB	3000	125	snap-0eeb064...	-	2025/11

Fault tolerance for all volumes in this Region

Snapshot summary

Recently backed up volumes / Total # volumes

0 / 3

Data Lifecycle Manager default policy for EBS Snapshots status

No default policy set up | Create policy

Last updated on Wed, Nov 19, 2025, 06:55:24 PM (GMT+05:30)

5. Assign passwordless authentication for the EC2 created in Task 2

I generated a new RSA SSH key on your local machine and copied the **public key**.
I connected to the EC2 instance using the **.pem** key.
Then you pasted your public key into the EC2 server's ~/.ssh/authorized_keys file.
Now you can log in **without a password or .pem**, using your own private SSH key.

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Nov 19 14:18:15 UTC 2025

System load: 0.0          Processes: 109
Usage of /: 29.9% of 6.71GB Users logged in: 0
Memory usage: 20%        IPv4 address for enX0: 172.31.17.156
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.
19 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Nov 19 14:13:57 2025 from 14.192.14.53
ubuntu@ip-172-31-17-156:~$ cd .ssh/
ubuntu@ip-172-31-17-156:~/.ssh$ vi authorized_keys
ubuntu@ip-172-31-17-156:~/.ssh$ exit
logout
Connection to ec2-52-207-226-14.compute-1.amazonaws.com closed.

ArjumandM@Arjumand MINGW64 ~/Downloads (master)
$ ssh ubuntu@ec2-52-207-226-14.compute-1.amazonaws.com
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Nov 19 14:19:33 UTC 2025

System load: 0.0          Processes: 109
Usage of /: 29.9% of 6.71GB Users logged in: 0
Memory usage: 20%        IPv4 address for enX0: 172.31.17.156
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.
19 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Nov 19 14:18:16 2025 from 14.192.14.53
ubuntu@ip-172-31-17-156:~$
```

6. Launch any EC2 using the spot purchasing option

Disable

Instance bandwidth configuration | Info

Select

Purchasing option | Info

☐ None

☐ Capacity Blocks
Launch instances for your active capacity blocks

☒ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

[Customize Spot instance options](#)

Capacity reservation | Info

Virtual server type (instance type)
t2.micro

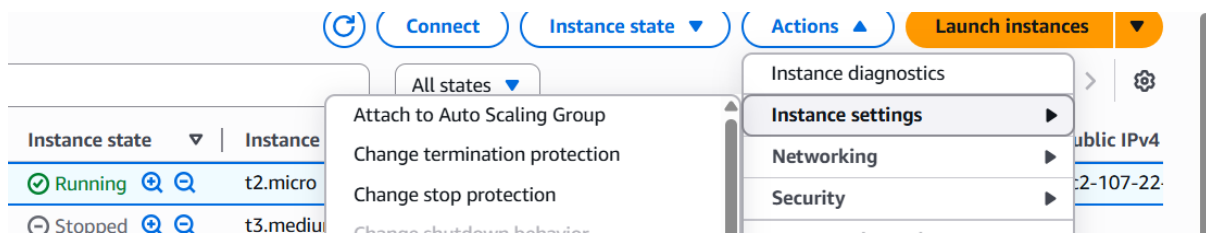
Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of o

Cancel

7. Enable termination policy on the EC2 created in Task 2



8. Launch one EC2 using AWS CLI.

Steps and Scripts You Performed

1 Launched a new EC2 instance using AWS CLI

You ran a CLI command to launch a new EC2 instance (Ubuntu/AMI ID + t2.micro) with your existing key pair, security group, subnet, and a tag.

Script you ran:

```
INSTANCE_ID=$(aws ec2 run-instances \
--image-id ami-0c02fb55956c7d316 \
--instance-type t2.micro \
--key-name Testing \
--security-group-ids sg-0c0e12b5987f18d09 \
--subnet-id subnet-0135156fd5cfd9836 \
--associate-public-ip-address \
--tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=cli-launched-
instance}]' \
--query 'Instances[0].InstanceId' --output text)
```

This command created the EC2 instance and stored the instance ID in the variable **INSTANCE_ID**.

2 Checked all instances to identify the correct instance ID

You listed all EC2 instance IDs in your account so you could confirm which one was created.

Script you ran:

```
aws ec2 describe-instances --query "Reservations[*].Instances[*].InstanceId" --output text
```


This helped you find the correct instance ID:

👉 **i-07ccfed9e5d3e2321**

3 Verified tags to ensure you selected the right instance

You checked the tags of that instance to make sure it was the one you launched via CLI.

Script you ran:

```
aws ec2 describe-instances --instance-id i-07ccfed9e5d3e2321 \  
--query "Reservations[0].Instances[0].Tags"
```

This returned:

```
{ "Key": "Name", "Value": "server_ubuntu" }
```

4 Enabled Termination Protection on the EC2 instance

This prevents the instance from being accidentally terminated.

Script you ran:

```
aws ec2 modify-instance-attribute \  
--instance-id i-07ccfed9e5d3e2321 \  
--disable-api-termination
```

aws

Search

[Alt+S]

EC2

>

Instances

EC2

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Dashboard

EC2 Global View

Events

Instances

i-0ccdf243343a07aba (Spot_server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

▼ Instance summary

Info

CloudShell

us-east-1

+

An error occurred (InvalidInstanceID.Malformed) when calling the DescribeInstances operation: Invalid id: "i-04db1c7c8bd8863"

~ \$ aws ec2 describe-instances --instance-id i-07ccfed9e5d3e2321 --query "Reservations[0].Instances[0].Tags"

[

{

"Key": "Name",

"Value": "server_ubuntu"

}

]

~ \$ aws ec2 modify-instance-attribute --instance-id <THE_CORRECT_ONE> --disable-api-termination

bash: THE_CORRECT_ONE: No such file or directory

~ \$ aws ec2 modify-instance-attribute --instance-id i-07ccfed9e5d3e2321 --disable-api-termination

~ \$ aws ec2 describe-instance-attribute --instance-id i-07ccfed9e5d3e2321 --attribute disableApiTermination

{

"DisableApiTermination": {

"Value": true

},

"InstanceId": "i-07ccfed9e5d3e2321"

}

~ \$